

Model Evaluation Report: Balanced vs. Imbalanced Training

1. Summary of Model Performance

Model	Test Set	Accuracy	Key Observations
Model_A (Balanced)	Balanced Test Set	0.473	Reasonable performance across all classes; fair distribution
Model_A (Balanced)	Imbalanced Test Set	0.568	Performs well on majority class (5star), lower on minority classes.
Model_B (Imbalanced)	Balanced Test Set	0.284	Bias towards 5 star; poor performance on minority classes
Model_B (Imbalanced)	Imbalanced Test Set	0.688	Performs well on majority class (5 star); minority classes underperform.

2.Observations

Balanced Model (Model_A):

- Maintains better fairness across all ratings.
- Accuracy drops slightly when applied to imbalanced test set because it hasn't seen as many majority-class reviews.

Imbalanced Model (Model_B):

- Strong performance on majority class (5*) in imbalanced test set.
- Poor generalization on balanced test set; overfits to the majority class.

Effect of training distribution:

- Balanced training ensures consistent performance across all classes.
- Imbalanced training favors majority class at the expense of minority classes.

3. Recommendation

- Deploy Model_A (Balanced) if fairness across ratings is important (real-world scenario where all ratings matter).
- Use Model_B (Imbalanced) only if most reviews are expected to be 5* and the goal is to optimize for majority-class accuracy.